



华中科技大学
Huazhong University of Science and Technology



华中科技大学
网络空间安全学院
School of Cyber Science and Engineering, HUST

实验一 玩转小乌龟

华中科技大学 2021 ROS 创新实践教学



日期：2021.03



内容：ROS进阶

0

概述

启动 小海龟



认识 结点

认识 服务

认识 话题



消息 操控小海龟



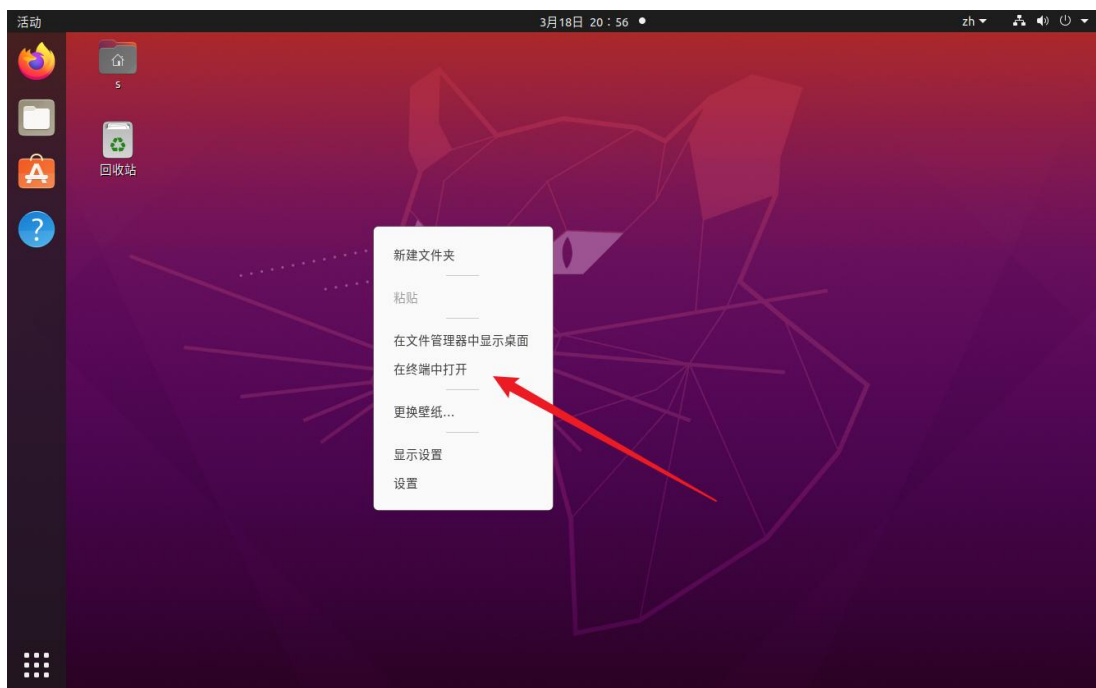
创建 功能包

本关涉及的指令：

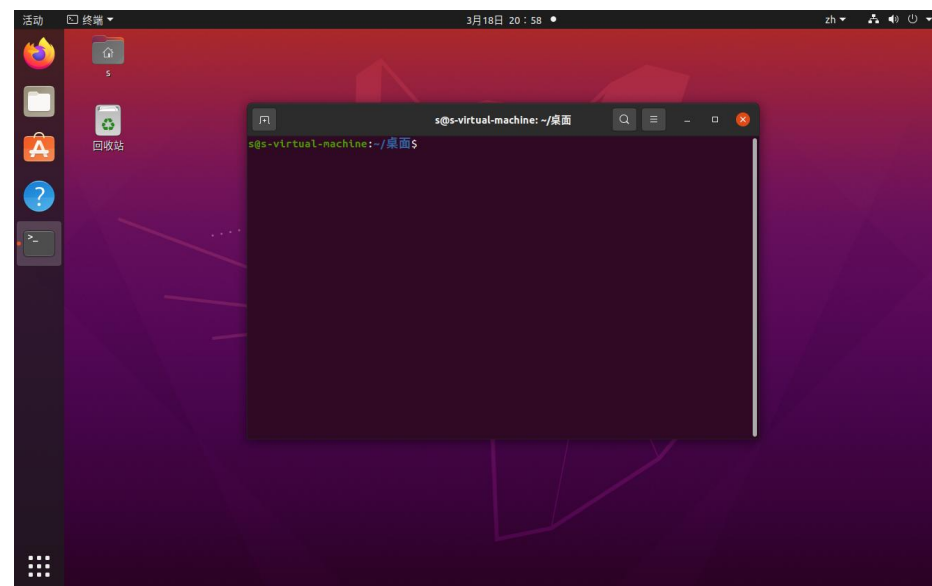
roscore：在新的终端中建立主服务器，

roslaunch turtlesim turtlesim_node：使用turtlesim包工具，建立Turtlesim节点

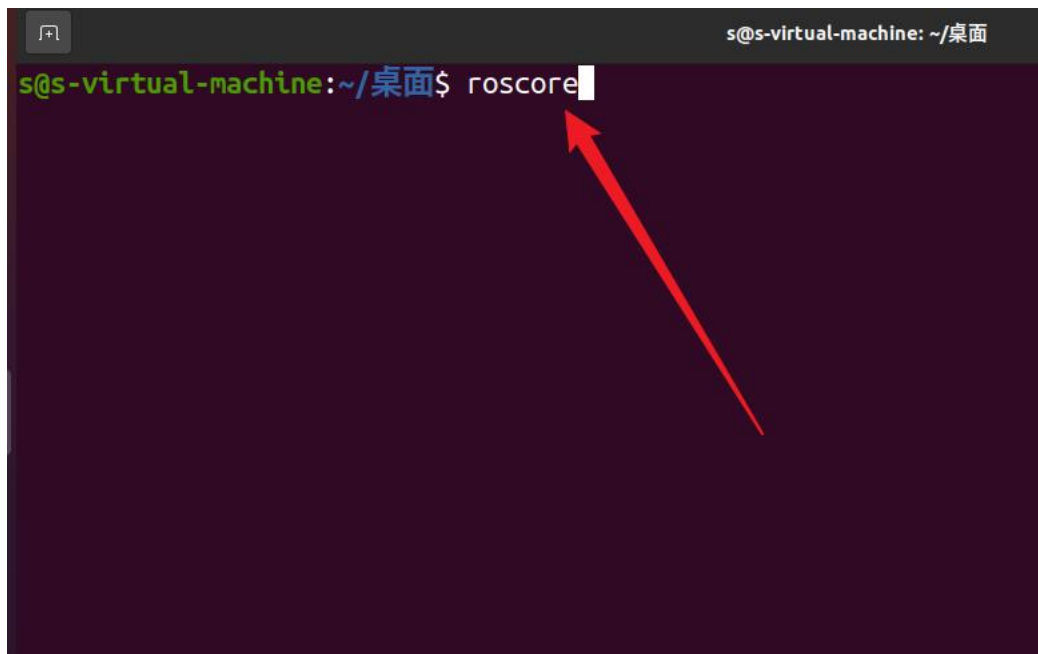
鼠标右键，点击在终端中打开
(或者使用快捷键ctrl+T，快速打开终端)



结果



输入命令roscore, 点击回车运行



A terminal window with a dark background. The title bar at the top reads 's@s-virtual-machine: ~/桌面'. The command prompt shows 's@s-virtual-machine:~/桌面\$' followed by the command 'roscore' and a white cursor. A red arrow points from the bottom right towards the end of the command 'roscore'.

```
s@s-virtual-machine:~/桌面$ roscore
```

roscore正常启动时

```
roscore http://localhost:11311/
shen@shen-virtual-machine:~/桌面$ roscore
... logging to /home/shen/.ros/log/f4ec9b58-03f8-11f0-b312-ef2dfa2e908a/rosc
og
Checking log directory for disk usage. This may take a while.
Press Ctrl-C to interrupt
Done checking log file disk usage. Usage is <1GB.

started roslaunch server http://localhost:34281/
ros_comm version 1.15.13

SUMMARY
=====

PARAMETERS
* /rostdistro: noetic
* /rosversion: 1.15.13

NODES
auto-starting new master
process[roscpp]: started with pid [5818]
ROS_MASTER_URI=http://localhost:11311/

setting /run_id to f4ec9b58-03f8-11f0-b312-ef2dfa2e908a
process[roscpp]: started with pid [5828]
started core service [/roscpp]
```

常见报错

```
shen@shen-virtual-machine:~/ros_catkin_ws$ roscore
... logging to /home/shen/.ros/log/44229c32-7649-11eb-8fe9-5b0df9286de4/roslaunch-shen-virtual-machine-2135.log
Checking log directory for disk usage. This may take a while.
Press Ctrl-C to interrupt
Done checking log file disk usage. Usage is <1GB.

RLError: Unable to contact my own server at [http://192.168.0.200:42939/].
This usually means that the network is not configured properly.

A common cause is that the machine cannot connect to itself. Please check
for errors by running:

    ping 192.168.0.200

For more tips, please see

    http://wiki.ros.org/ROS/NetworkSetup

The traceback for the exception was written to the log file
```

解决方式

```
shen@shen-virtual-machine:~/桌面$ gedit ~/.bashrc
shen@shen-virtual-machine:~/桌面$ source ~/.bashrc
shen@shen-virtual-machine:~/桌面$

120 export ROS_MASTER_URI=http://localhost:11311
121 export ROS_HOSTNAME=localhost
122 export ROS_IP='hostname -I'
```

点击右上角唤出新终端



在新终端中执行
`roslaunch turtlesim turtlesim_node`
命令

```
s@s-virtual-machine: ~/桌面
roslaunch http://localhost:11311/
s@s-virtual-machine: ~/桌面
s@s-virtual-machine:~/桌面$ roslaunch turtlesim turtlesim_node
[ INFO] [1742303415.280088196]: Starting turtlesim with node name /turtlesim
[ INFO] [1742303415.286477814]: Spawning turtle [turtle1] at x=[5.544445], y=[5.544445]
0]
```

结果



不要关闭这两个终端或杀死其中的进程，
接下来可以打开新的终端查看节点信息，控制
小乌龟运动了。

本关涉及的指令：

rostopic list：列出系统中已发布的节点

rostopic info /turtlesim：调出/turtlesim结点的信息

需要解决的问题：

Q1：调出/turtlesim的信息，并分析信息的含义。

rostopic list

```
shen@sh... × shen@sh... × shen@sh... × shen@sh... ×  
shen@shen-virtual-machine:~/桌面$ rostopic list  
/rosout  
/turtlesim
```

/rosout

默认结点

/turtlesim

新建的结点

rostopic info /turtlesim

```
shen@shen-virtual-machine:~/桌面$ rostopic info /turtlesim
-----
Node [/turtlesim]
Publications:
* /rosout [roscpp_msgs/Log]
* /turtle1/color_sensor [turtlesim/Color]
* /turtle1/pose [turtlesim/Pose]

Subscriptions:
* /turtle1/cmd_vel [unknown type]
```

Services:

```
* /clear
* /kill
* /reset
* /spawn
* /turtle1/set_pen
* /turtle1/teleport_absolute
* /turtle1/teleport_relative
* /turtlesim/get_loggers
* /turtlesim/set_logger_level
```

/Services

Connect

contacting node http://localhost:43523/ ...

Pid: 4542

Connections:

```
* topic: /rosout
  * to: /rosout
  * direction: outbound (38319 - 127.0.0.1:34414) [23]
  * transport: TCPROS
```

本关涉及的指令：

rosservice list: 列出服务名称

rosservice call /spawn [Tab]: 在同一个窗口中创建一个新的小乌龟

需要解决的问题：

Q2: /spawn名为学号的小乌龟，使用rosservice call /teleop_* 控制其运动，并完成 /kill，记录每一步参数的并解释其含义。

rosservice list

```
shen@shen-virtual-machine:~/桌面$ rosservice list  
/clear  
/kill  
/reset  
/rosout/get_loggers  
/rosout/set_logger_level  
/spawn ←  
/turtle1/set_pen  
/turtle1/teleport_absolute  
/turtle1/teleport_relative  
/turtlesim/get_loggers  
/turtlesim/set_logger_level
```

可以通过名称了解
service的功能。
例如：spawn产卵

rosservice call /spawn [Tab][Tab]

```
shen@shen-virtual-machine:~/桌面$ rosservice call  
/spawn "x: 3.0  
y: 5.0  
theta: 0.0  
name: 'shen'  
name: "shen"
```

x,y,theta等可手动输入，**注意空格**

也可用[Tab][Tab]补全

结果



本关涉及的指令：

rostopic list: 了解一下turtlesim包含的所有话题

rostopic info /turtle1/pose: 了解位姿的话题的具体信息

rostopic type /turtle1/pose: 了解位姿的话题的数据类型

rosmmsg show turtlesim/Pose: 了解位姿的话题的数据结构

rostopic echo /turtle1/pose: 对turtlesim/Pose 话题进行实时监听

需要解决的问题：

Q3: 说明/cmd_vel 的连接信息、传送数据类型、数据类型的数据结构。

```
shen@shen-virtual-machine:~/A/src/learning_topic/scripts$ rostopic
list
/rosout
/rosout_agg
/turtle1/cmd_vel
/turtle1/color_sensor
/turtle1/pose
shen@shen-virtual-machine:~/A/src/learning_topic/scripts$ rostopic
info /turtle1/pose
Type: turtlesim/Pose

Publishers:
* /turtlesim (http://localhost:33813/)

Subscribers: None

shen@shen-virtual-machine:~/A/src/learning_topic/scripts$ rostopic
type /turtle1/pose
turtlesim/Pose
shen@shen-virtual-machine:~/A/src/learning_topic/scripts$ rosmg s
how turtlesim/Pose
float32 x
float32 y
float32 theta
float32 linear_velocity
float32 angular_velocity
```

列出话题

显示话题pose的信息

显示话题pose的数据类型

显示话题pose的数据结构

本关涉及的指令：

rostopic type /turtle1/cmd_vel: 查看与移动有关的话题的信息

rostopic pub -1 /turtle1/cmd_vel geometry_msgs/Twist [Tab][Tab]: 发送控制消息

rosservice call /reset: 重启窗口（还原初始状态）

需要解决的问题：

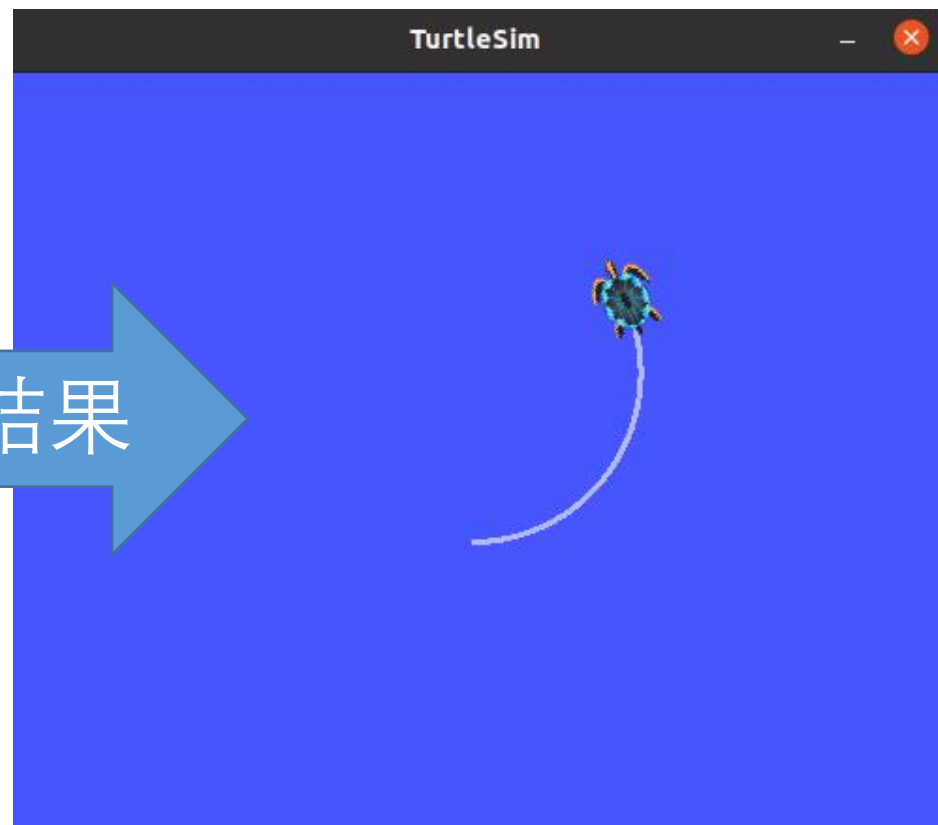
Q4：发送命令使乌龟运动。

要求：以1Hz的频率连续发送命令 线速度4.0 右旋角速度2.0

**rostopic type /turtle1/cmd_vel,
rostopic pub -1 /turtle1/cmd_vel
geometry_msgs/Twist [Tab][Tab]**

```
shen@shen-virtual-machine:~/A/src/learning_topic/scripts$ rostopic  
type /turtle1/cmd_vel  
geometry_msgs/Twist  
shen@shen-virtual-machine:~/A/src/learning_topic/scripts$ rostopic  
pub -1 /turtle1/cmd_vel geometry_msgs/Twist "linear:  
x: 4.0  
y: 0.0  
z: 0.0  
angular:  
x: 0.0  
y: 0.0  
z: 2.0"  
publishing and latching message for 3.0 seconds
```

结果



本关涉及的指令：

catkin_make

catkin_create_pkg

rqt_plot /turtle1/pose/x:y

roslaunch learning_topic turtlesim.py

创建功能包的指令见指导书。

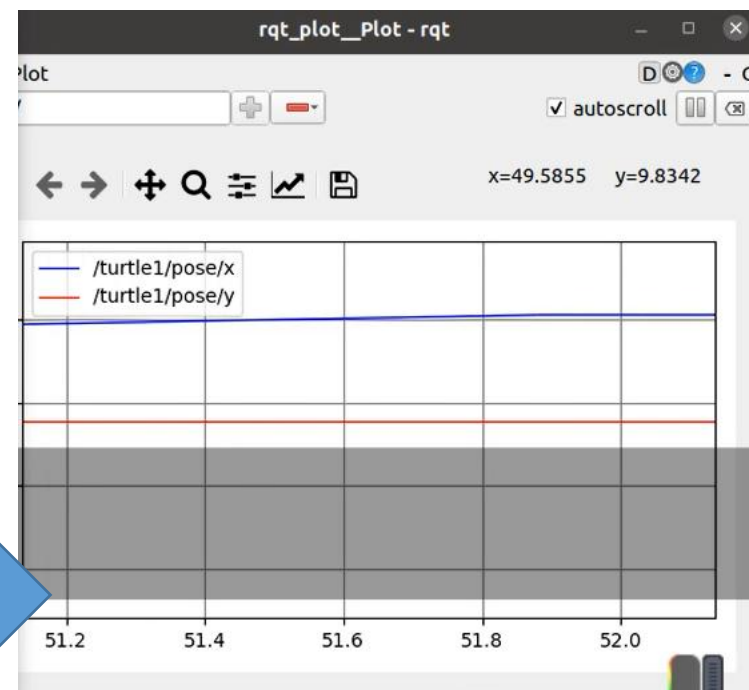
`rqt_plot /turtle1/pose/x:y`

`roslaunch learning_topic turtlesim.py`

```
shen@sh... x  roscore ... x  shen@sh... x  shen@sh... x
shen@shen-virtual-machine:~/桌面$ rqt_plot /turtle
```

```
shen@sh... x  roscore ... x  shen@sh... x  shen@sh... x
shen@shen-virtual-machine:~/桌面$ roslaunch learning_
topic turtlesim.py
[INFO] [1615614925.616614]: Press CTRL+c to stop
TurtleBot
[INFO] [1615614925.619380]: Set rate 10Hz
^C[INFO] [1615614934.460899]: Stopping Turtlesim
```

结果



The background features a large, light blue downward-pointing triangle. Overlaid on this is a thin, light blue upward-pointing triangle. A thick, dark blue arrow points from the upper left towards the center, and another thick, dark blue arrow points downwards from the center. The text "THANK YOU" is centered in a bold, dark blue, sans-serif font.

THANK YOU